INCH POUND

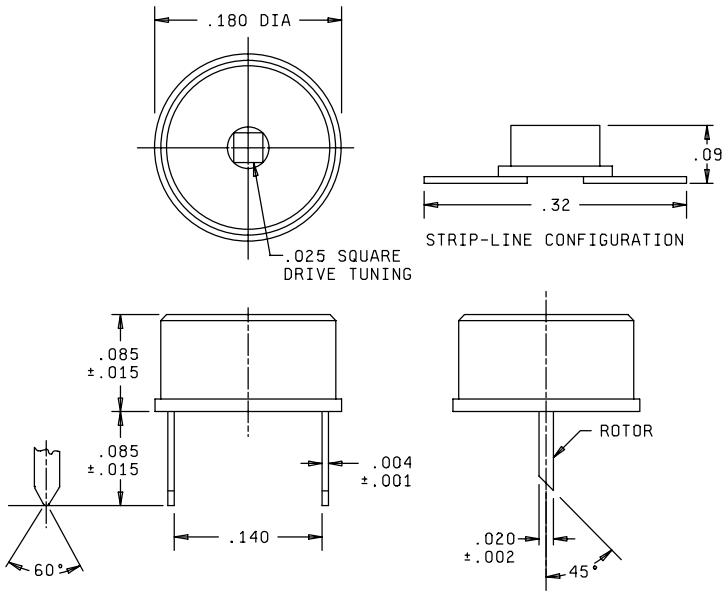
MIL-PRF-81/9C
25 May 1999
SUPERSEDING
MIL-C-81/9B
27 December 1989

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, VARIABLE, CERAMIC DIELECTRIC,
STYLE CV98

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-81.




| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .025 | 0.64 |
| .002 | 0.05 | .085 | 2.16 |
| .004 | 0.10 | .090 | 2.29 |
| .005 | 0.13 | .140 | 3.56 |
| .015 | 0.40 | .180 | 4.57 |
| .020 | 0.51 | .320 | 8.13 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 1. Dimensions and configurations.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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Capacitance value: See table I.

DC rated voltage: 250 V dc.

Operating temperature range: -55°C to +125°C.

Characteristics: See table I.

Dielectric withstanding voltage: Method 301 of MIL-STD-202. A dc potential of 500 V dc applied between terminals for 3 seconds ±2 seconds.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 feet).

Test potential: 100 percent of dc rated voltage.

Insulation resistance: Method 302 of MIL-STD-202, condition A, 100 V dc applied: 10,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.

DF: At 1 MHz ±100 kHz, at maximum and minimum capacitance: Shall be not more than 0.2 percent.

TABLE I. Style CV98.

| Type designation 1/ | Capacitance | | DC rated voltage | Symbol | Characteristic | Minimum Q at 100 MHz |
|---|---|---|---|---|---|---|
| | Min (pF) | Max (pF) | | | Temperature coefficient | |
| CV98F2R5-- | 0.5 | 2.5 | 250 | F | +100 ±150 | 1,000 |
| CV98W5R0-- 2/ | 1.0 | 5.0 | 250 | W | 0 ±400 | 1,000 |
| CV98H100-- | 2.5 | 10.0 | 250 | H | +100 ±600 | 1,000 |
| CV98Q250-- | 6.0 | 25.0 | 250 | Q | -1,500 ±900 | 300 |
| CV98J150-- | 5.0 | 15.0 | 250 | J | -300 ±300 | 750 |
| CV98K200-- | 5.0 | 20.0 | 250 | K | -450 ±300 | 500 |

1/ For strip-line configuration parts, the complete type designation shall include additional symbols SL.
2/ Type designation CV98W5R0-- is substitutable for type designator CV98G5R0--.

Temperature coefficient: Within the limits specified for the applicable characteristics.

Capacitance drift: Within 0.75 percent of initial step 1 measurement or 0.50 picofarad (pF), whichever is greater.

Terminal strength:

Pull test: Capacitor held by body and 1-pound load.

Torque: Not less than 0.05 ounce-inch nor more than 1.0 ounce-inch.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 g's).



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Vibration, high frequency: Method 204 of MIL-STD-202, condition B (15 g's).

Capacitance change: Shall not exceed ±2 percent or 0.5 pF, whichever is greater.

DF: Shall be not more than 0.2 percent.

Dielectric withstanding voltage: 500 V dc, applied for 3 seconds ±2 seconds.

Insulation resistance: Shall be 10,000 megohms, minimum.

Fatigue:

$\Delta C$: Shall not exceed 12 percent or 0.75 pF, whichever is greater.

Torque: Initial requirement.

Life:

Qualification test: 1,000 hours at +85°C, 150 percent of rated V dc applied.

Insulation resistance: Initial requirement.

Capacitance change: Shall not exceed ±8 percent of initial value or 0.5 pF, whichever is greater.

Group C life: Conditions and requirements are the same as that for qualification except that capacitance change shall not exceed ±5 percent of initial value or 0.5 pF, whichever is greater.

Moisture resistance: Method 106 of MIL-STD-202.

Insulation resistance: Shall be 10,000 megohms minimum.

Capacitance change: Shall not exceed ±5 percent of nominal value or 0.5 pF, whichever is greater.

DF: Shall be not more than 0.5 percent.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Air Force - 19, 99

Preparing activity:
DLA - CC

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